
Lecture 5: Applications of Diffusion

Chenyang Yuan

January 14, 2026



Outline

- Announcements + Recap
- Applications
 - Latent diffusion, video generation and editing
 - Imitation learning in robotics
- Why does diffusion work so well?
 - The power of iterative denoising/refinement
 - Diffusion as minimizing distance to data manifold



Recap: Perceptually important freqs

Which noise levels should we focus training on?

Adding small amounts of noise does not significantly affect perceptual quality

No need to focus training on small noise levels

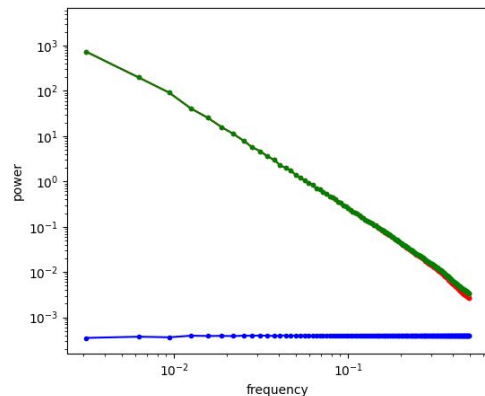


Image credit: Diffusion is spectral autoregression <https://sander.ai/2024/09/02/spectral-autoregression.html>



How to focus on different frequencies?

1. Modifying noise schedule

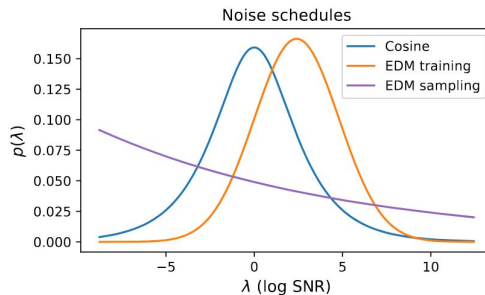
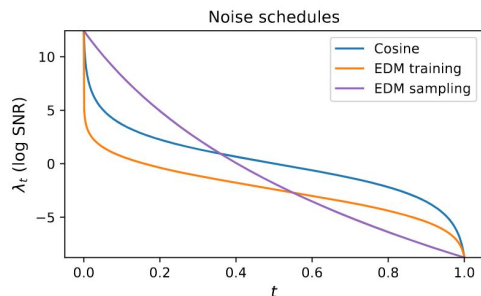


Image credit: Understanding Diffusion Objectives as the ELBO with Simple Data Augmentation <https://arxiv.org/pdf/2303.00848>

2. Modifying training objective

What to predict when training?

$$x_\sigma = x_0 + \sigma \epsilon$$

$$\text{Noise} \quad L_\epsilon(\theta) = \mathbf{E}_{x_0, \sigma, \epsilon} \|\epsilon_\theta(x_\sigma, \sigma) - \epsilon\|^2$$

$$\text{Clean data} \quad L_x(\theta) = \mathbf{E}_{x_0, \sigma, \epsilon} \|x_\theta(x_\sigma, \sigma) - x_0\|^2$$

$$\text{Rectified flow} \quad L_v(\theta) = \mathbf{E}_{x_0, \sigma, \epsilon} \|v_\theta(x_\sigma, \sigma) - (\epsilon - x_0)\|^2$$

Mathematically equivalent for fixed σ , but reweighs loss by a function of σ !

$$\begin{aligned} L_x(\theta) &= \mathbf{E} \|x_\theta(x_\sigma, \sigma) - x_0\|^2 = \mathbf{E} \left[\sigma^2 \left\| \frac{x_\theta(x_\sigma, \sigma) - x_\sigma}{\sigma} - \frac{x_0 - x_\sigma}{\sigma} \right\|^2 \right] \\ &= \mathbf{E} \left[\sigma^2 \|\epsilon_\theta(x_\sigma, \sigma) - \epsilon\|^2 \right] \end{aligned}$$

High-resolution image generation

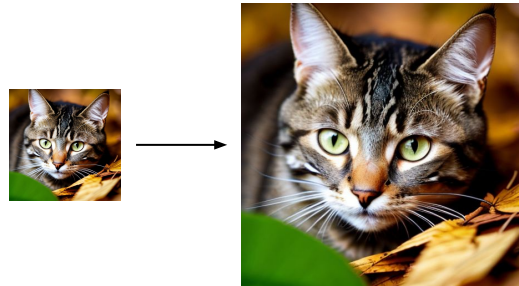


How to generate high-resolution images?

Naive approach: Make denoising U-net larger (e.g. 64x64 to 512x512)

Pixel-space an inefficient representation of detail

How to compress perceptually unimportant frequencies?

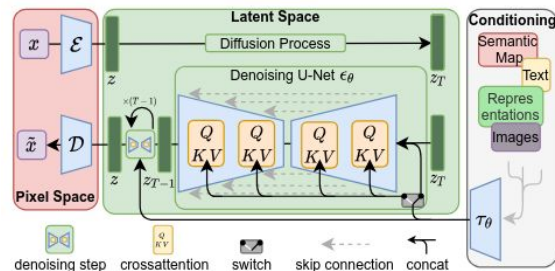


High-Resolution Image Synthesis with Latent Diffusion Models

Robin Rombach¹ * Andreas Blattmann¹ * Dominik Lorenz¹ Patrick Esser² Björn Ommer¹

¹Ludwig Maximilian University of Munich & IWR, Heidelberg University, Germany ²Runway ML

<https://github.com/CompVis/latent-diffusion>



Latent diffusion training

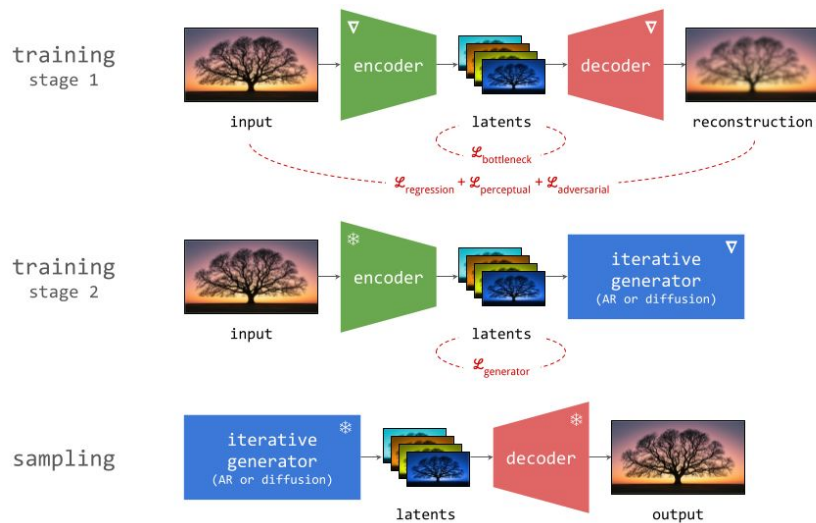


Image credit: Generative modelling in latent space <https://sander.ai/2025/04/15/latents.html>



512 x 512 x 3



64 x 64 x 4



512 x 512 x 3



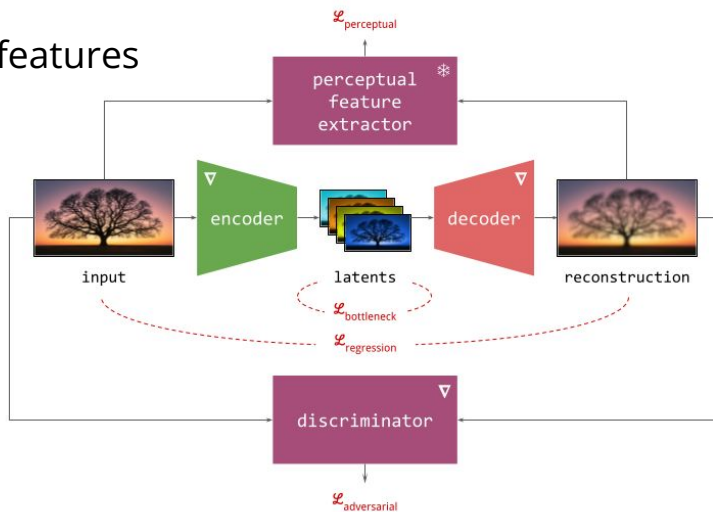
VAE training

Perceptual: Focus on visually important features

Regression: MSE in pixel space

Adversarial: Improve realism (i.e. GAN)

Bottleneck: Constrain latent capacity



Powerful idea that extends to many other domains!

Image credit: Generative modelling in latent space <https://sander.ai/2025/04/15/latents.html>

Why diffuse on latent representations?

Preserve spatial structure while compressing perceptually insignificant details

Although spatial structure introduces inefficiencies in latent representation

End-to-end training v.s. Two-stage training?

Diffusion in latent space builds-in perceptual priors for efficiency

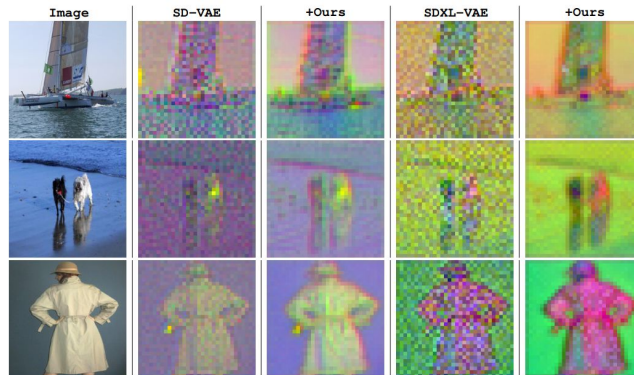


Image credit: EQ-VAE <https://arxiv.org/pdf/2502.09509>



Further applications of image diffusion models



Text-to-video generation



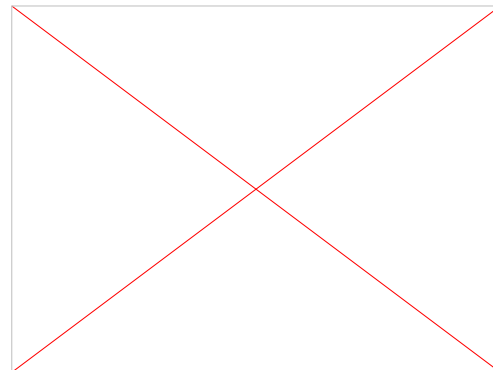
A girl is running across a beach and holding a kite. She's wearing jean shorts and a yellow t-shirt. The sun is shining down.

Video credit: <https://ai.meta.com/research/movie-gen/>



Low-angle tracking shot, 18mm lens. The car drifts, leaving trails of light and tire smoke, creating a visually striking and abstract composition. The camera tracks low, capturing the sleek, olive green muscle car as it approaches a corner. As the car executes a dramatic drift, the shot becomes more stylized. The spinning wheels and billowing tire smoke, illuminated by the surrounding city lights and lens flare, create streaks of light and color against the dark asphalt. The cityscape – yellow cabs, neon signs, and pedestrians – becomes a blurred, abstract backdrop. Volumetric lighting adds depth and atmosphere, transforming the scene into a visually striking composition of motion, light, and urban energy.

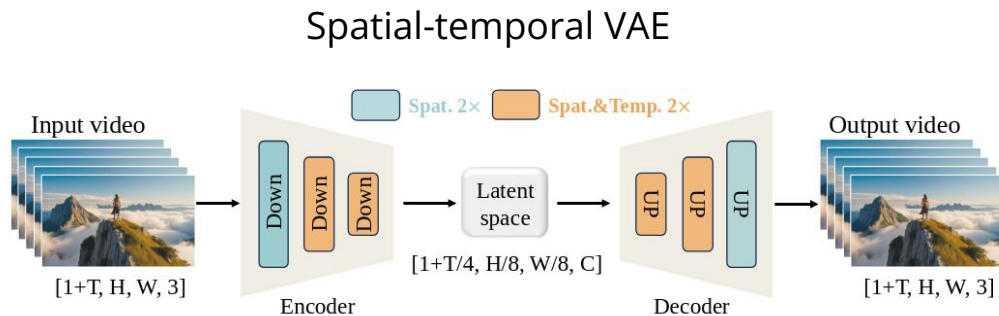
Video credit: <https://deepmind.google/technologies/veo/veo-2/>



In an ornate, historical hall, a massive tidal wave peaks and begins to crash. Two surfers, seizing the moment, skillfully navigate the face of the wave.

Video credit: <https://openai.com/index/video-generation-models-as-world-simulators/>

How does video generation work?



Challenges:

1. Temporal consistency and how to best perceptually compress motion
2. Autoregressive generation of arbitrary-length videos

Image credit: WAN 2.1 <https://arxiv.org/pdf/2503.20314>

Image editing and composition



Figure 1. With just a few images (typically 3-5) of a subject (left), *DreamBooth*—our AI-powered photo booth—can generate a myriad of images of the subject in different contexts (right), using the guidance of a text prompt. The results exhibit natural interactions with the environment, as well as novel articulations and variation in lighting conditions, all while maintaining high fidelity to the key visual features of the subject.

Image credit: DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation <https://arxiv.org/pdf/2208.12242>

Generating visual illusions

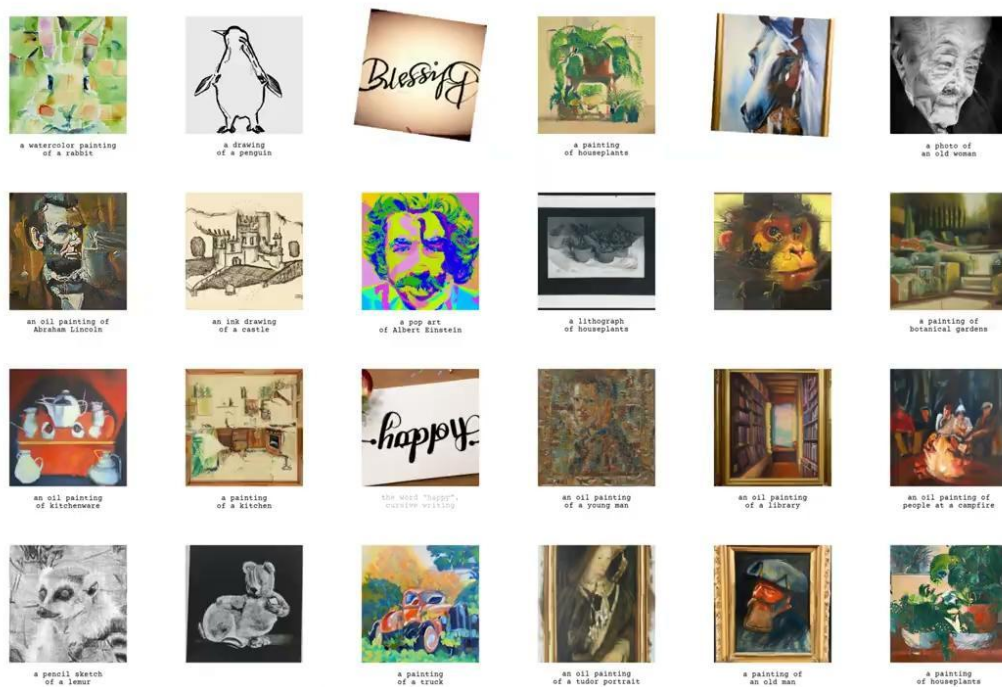


Image credit: Visual anagrams https://dangeng.github.io/visual_anagrams/



Fine-grained control of diffusion models



Figure 1: Controlling Stable Diffusion with learned conditions. ControlNet allows users to add conditions like Canny edges (top), human pose (bottom), *etc.*, to control the image generation of large pretrained diffusion models. The default results use the prompt “a high-quality, detailed, and professional image”. Users can optionally give prompts like the “chef in kitchen”.

Image credit: Adding Conditional Control to Text-to-Image Diffusion Models <https://arxiv.org/pdf/2302.05543>

ControlNet

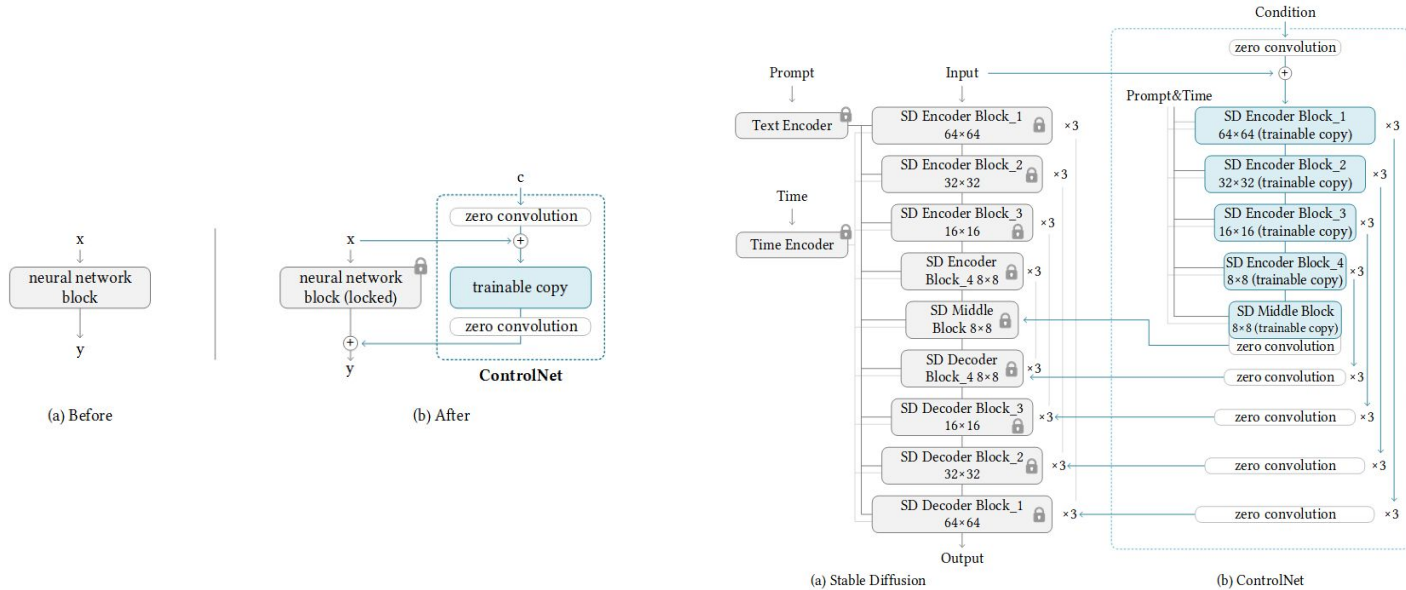
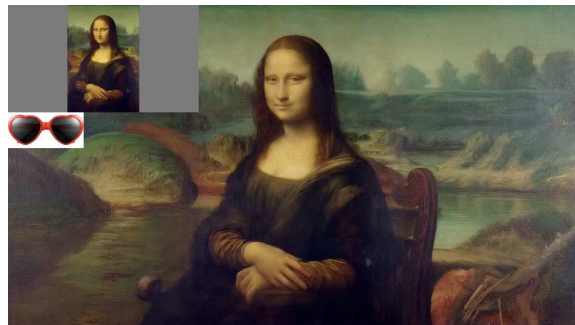
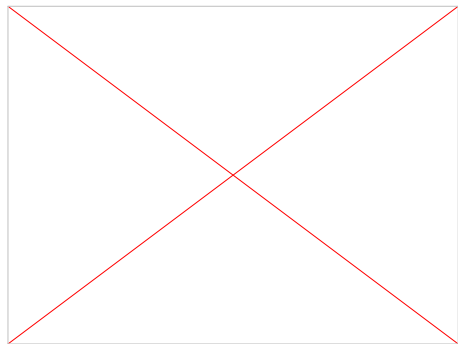


Image credit: Adding Conditional Control to Text-to-Image Diffusion Models <https://arxiv.org/pdf/2302.05543>

Video Editing



Video credit: VACE: All-in-One Video Creation and Editing <https://arxiv.org/pdf/2503.07598>



Massachusetts Institute of Technology

6.S183 IAP 2026: A practical introduction to diffusion models

Video game simulation



Image/video credit: Diffusion Models Are Real-Time Game Engines
<https://gamengen.github.io/>

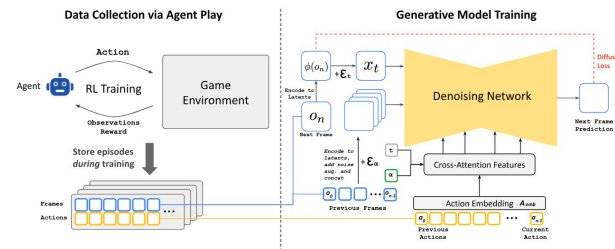


Figure 3: GameNGen method overview. v-prediction details are omitted for brevity.

World models



Video credit: <https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>



Massachusetts Institute of Technology

6.S183 IAP 2026: A practical introduction to diffusion models

Diffusion Models in Robotics



Why is robotics hard?

Traditional control systems need precise dynamics models

Interaction with the real physical world

Moravec's paradox:

"It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility"

Video credit: IEEE Spectrum



Massachusetts Institute of Technology

6.S183 IAP 2026: A practical introduction to diffusion models

Which tasks are harder than others?



Entering a round-knob push door



Entering a lever-handle self-closing push door



Entering a lever-handle self-closing pull door



Video credit: Benjie's Humanoid Olympic Games
<https://generalrobots.substack.com/p/benjie-humanoid-olympic-games>



Massachusetts Institute of Technology

6.S183 IAP 2026: A practical introduction to diffusion models

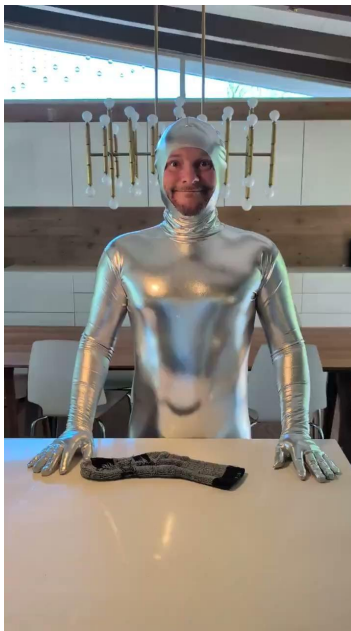
Which tasks are harder than others?



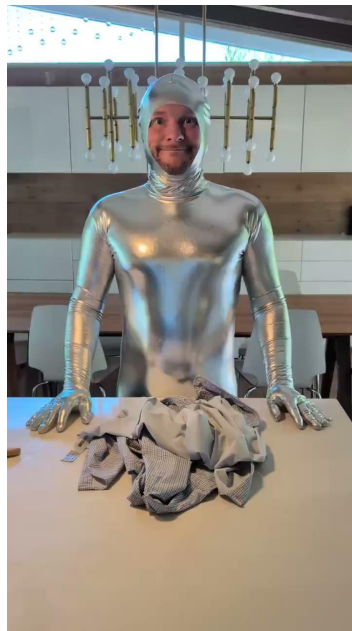
Fold an inside-out
T-shirt



Turn a sock inside-out



Hang a men's dress-shirt



Video credit: Benjie's Humanoid Olympic Games
<https://generalrobots.substack.com/p/benjies-humanoid-olympic-games>



Massachusetts Institute of Technology

6.S183 IAP 2026: A practical introduction to diffusion models

Imitation learning in robotics



Video credit: <https://toyotaresearchinstitute.github.io/lbm1/>

Diffusion policy

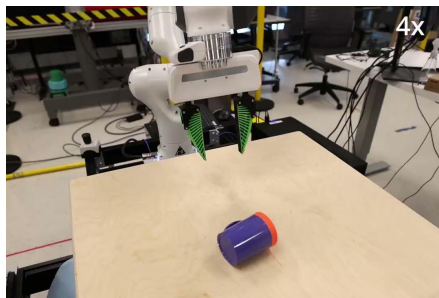
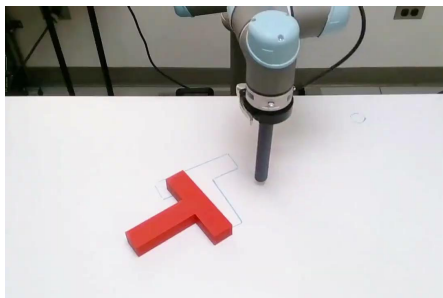
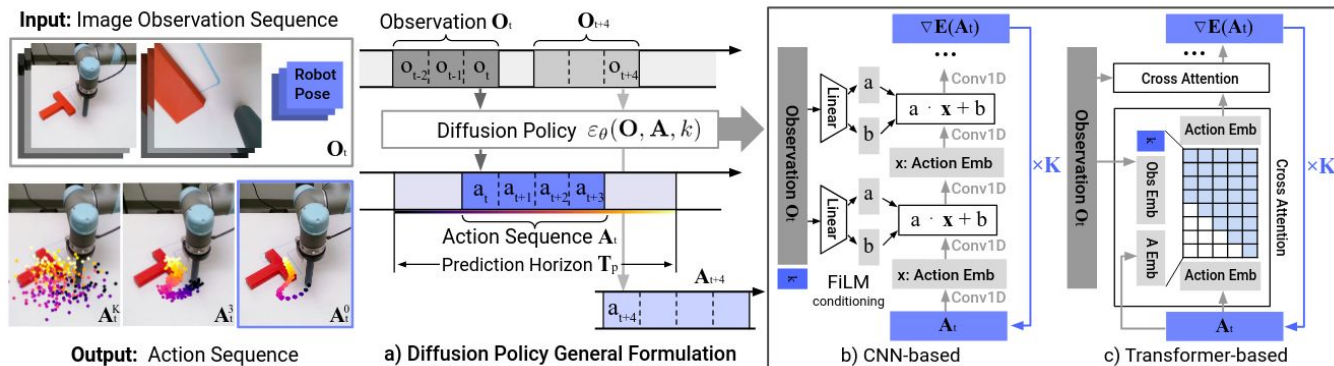


Image and video credit:
<https://diffusion-policy.cs.columbia.edu/>

Large Behavioral Models



Video credit: <https://www.pi.website/blog/olympics>



Video credit: Boston Dynamics

Why does diffusion policy work?

1. Architectural improvements borrowed from vision models
2. Noise injection during training to improve learning dynamics
3. Supervised iterative compute to generate output via many denoising steps



Much Ado About Noising: Dispelling the Myths of Generative Robotic Control:
<https://simchowitzlabpublic.github.io/much-ado-about-noising-project/>

Video credit: <https://www.pi.website/blog/olympics>



Optimization interpretation of diffusion

The *distance function* $\text{dist}_{\mathcal{K}} : \mathbb{R}^n \rightarrow \mathbb{R}$ to a set $\mathcal{K} \subseteq \mathbb{R}^n$, is defined via

$$\text{dist}_{\mathcal{K}}(x) := \inf\{\|x - x_0\| : x_0 \in \mathcal{K}\}$$

The *projection* of $x \in \mathbb{R}^n$, is the set of points that attain this distance

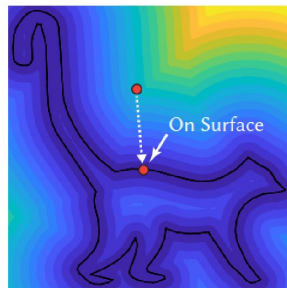
$$\text{proj}_{\mathcal{K}}(x) := \{x_0 \in \mathcal{K} : \text{dist}_{\mathcal{K}}(x) = \|x - x_0\|\}$$

Intuitively, $\text{proj}_{\mathcal{K}}(x) - x$ is the direction of steepest descent (i.e. neg. gradient) of $\text{dist}_{\mathcal{K}}(x)$.

Proposition

Suppose $\mathcal{K} \subseteq \mathbb{R}^n$ is closed and $x \notin \mathcal{K}$. If $\text{proj}_{\mathcal{K}}(x)$ is a singleton, then

$$\nabla \frac{1}{2} \text{dist}_{\mathcal{K}}(x)^2 = x - \text{proj}_{\mathcal{K}}(x)$$



Interpreting and Improving Diffusion Models from an Optimization Perspective <https://arxiv.org/abs/2306.04848>

Diffusion as gradient descent on distance function

Ideal denoiser equivalent to gradient of smoothed distance function

Theorem

For all $\sigma > 0$ and $x \in \mathbb{R}^n$, we have $\frac{1}{2} \nabla_x \text{dist}_{\mathcal{K}}^2(x, \sigma) = \sigma \epsilon^*(x, \sigma)$.

Interpretation: Diffusion aims to minimize distance to data manifold $\mathcal{K}$

Deterministic (DDIM) update: $x_{t-1} = x_t + (\sigma_{t-1} - \sigma_t) \epsilon_{\theta}(x_t, \sigma_t)$

Let $f(x) := \frac{1}{2} \text{dist}_{\mathcal{K}}^2(x)$. If $\sigma \approx \text{dist}_{\mathcal{K}}(x)/\sqrt{n}$, then $\nabla f(x) \approx \sigma \epsilon_{\theta}(x)$

DDIM is approximate gradient descent on f with stepsize $1 - \frac{\sigma_{t-1}}{\sigma_t}$, with $\nabla f(x_t)$ estimated by $\epsilon_{\theta}(x_t, \sigma_t)$

Interpreting and Improving Diffusion Models from an Optimization Perspective <https://arxiv.org/abs/2306.04848>

Manifold adherence

Noise injection and supervised iterative compute allow for better manifold adherence given out-of-distribution observations

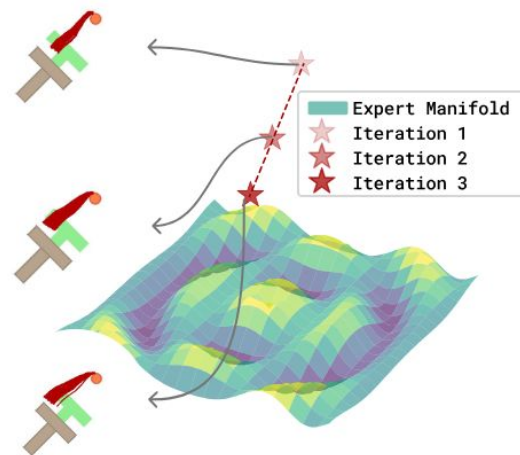


Image credit: Much Ado About Noising: Dispelling the Myths of Generative Robotic Control: <https://arxiv.org/pdf/2512.01809>

Feedback/Attendance Survey



<https://forms.gle/yrLAXm9KyYiEB6SQ9>



Massachusetts Institute of Technology

6.S183 IAP 2026: A practical introduction to diffusion models

Additional readings

[Generative modelling in latent space](#)

[Underactuated robotics class notes on imitation learning](#)

[Much Ado About Noising: Dispelling the Myths of Generative Robotic Control](#)

[Interpreting and Improving Diffusion Models from an Optimization Perspective](#)